

The ethics of uncertain sentience is an area of applied ethics concerned with decision-making when it is unclear whether a being is sentient, meaning capable of subjective experience, feeling, or perception. The issue appears in animal ethics, especially for invertebrates such as crustaceans, cephalopods, and insects, and for fish, where the extent of their capacity to experience pain is disputed. It also appears in environmental ethics, the ethics of artificial intelligence and neuroethics.

Proposed responses include the precautionary principle, incautious and expected-value approaches, virtue-ethical arguments for attentiveness toward possibly sentient animals, and assessment frameworks in animal welfare science. Discussion also concerns evidential standards, regulatory and economic costs, and the scope of moral consideration across biological taxa and computational substrates.

Animal ethics

Invertebrates

Crustaceans and cephalopods

David Foster Wallace's 2004 essay "Consider the Lobster" describes the Maine Lobster Festival and discusses the difficulty of inferring pain across species. It notes evidence of nociceptors in lobsters alongside uncertainty about endogenous opioids, and considers the ethics of killing and cooking animals alive. Robert C. Jones's chapter "The Lobster Considered" engages Wallace's essay and argues that the available evidence supports treating lobsters as capable of pain and therefore as morally considerable. Jones reviews neurophysiological and behavioural work on nociception and opioid systems, distinguishes moral considerability from degrees of moral significance, and concludes that a precautionary approach is warranted toward practices that risk causing pain to crustaceans.

A 2021 UK government-commissioned review by the London School of Economics evaluated 300 studies and concluded that cephalopods and decapod crustaceans should be treated as

sentient. It graded the evidence as "very strong" for octopods, "strong" for most crabs, and "substantial but not strong" for squid, cuttlefish, and lobsters. The review recommended best-practice transport, stunning and slaughter, and said lobsters and crabs should not be boiled alive. The review informed debate on the Animal Welfare (Sentience) Act.

Insects

In 2016, Shelley A. Adamo reviewed philosophical, neurobiological, behavioural, robotic and evolutionary evidence on insect pain and concluded that the question remains unsettled. Adamo writes that insects show nociception and complex learning, but that similar pain-like behaviours can arise from simpler mechanisms and can be engineered in robots. She argues that argument by analogy to human pain is weak without a clear account of the neural architecture needed for subjective experience. She contrasts Morgan's canon with the precautionary principle, noting that they point to opposing policy responses and that precaution has research and economic costs. She nonetheless recommends careful handling to avoid stress for methodological and ethical reasons.

In an article for Vox, Dylan Matthews discusses insect sentience in proposals to scale entomophagy. He reports limited evidence on whether farmed insects feel pain and on the welfare effects of common slaughter methods, including freezing and shredding. He cites estimates that around 1 trillion insects are raised and killed annually, with about 79–94 billion alive at any time, and argues that if insects can suffer, the ethical implications of expanding insect farming would be large.

Fish

Maximilian Padden Elder argues that contemporary evidence warrants treating fish as potential sufferers within animal ethics. He distinguishes nociception from conscious pain and contends that teleosts possess nociceptors and display behaviours consistent with affective

states. He describes objections based on the absence of a neocortex, or on the absence of human-like pain displays, as anthropocentric. Elder cites subcortical processing and behavioural data against neocortex-based dismissals and cautions against using human responses as the standard for other species. He also discusses cultural and psychological factors that reduce empathy for fish and lower concern for their welfare.

Given remaining uncertainty, Elder advocates a precautionary approach that shifts the burden of proof to those whose actions risk harm. He cites policy analogues including the US Marine Mammal Protection Act, UK protection of cephalopods, and European Union uses of the precautionary principle. He also points to scale as a reason for priority, noting estimates of roughly 1–2.7 trillion wild-caught fish annually and tens of billions of farmed fish slaughtered in a single year. He discusses implications for commercial and recreational fishing and questions moral pescetarianism in light of possible fish suffering.

Decision principles and frameworks

In the 2015 essay "Reconsider the Lobster", Jeff Sebo quotes Wallace's discussion of the difficulty of establishing whether an animal can experience pain. Sebo calls the question of how to treat individuals of uncertain sentience the "sentience problem" and argues that this problem, which "Wallace raises deserves much more philosophical attention than it currently receives". Sebo identifies two assumptions behind the problem: "sentientism about moral status", the view that sentient individuals deserve moral consideration, and "uncertainty about other minds", the scientific and philosophical uncertainty about which individuals are sentient. Sebo discusses three approaches. The incautionary principle holds that in cases of uncertainty about sentience it is morally permissible to treat individuals as if they are not sentient. The precautionary principle holds that in such cases there is a moral obligation to treat them as if they are sentient. The expected value principle holds that people are "morally required to multiply our credence that they are by the amount of moral value they would have if they

were, and to treat the product of this equation as the amount of moral value that they actually have". Sebo advocates the expected-value approach.

Philosopher Jonathan Birch proposes a practical framework grounded in the precautionary principle for assessing animal sentience and argues that it is consistent with established practice in animal welfare science.

Simon Knutsson and Christian Munthe argue from the perspective of virtue ethics that, in relation to animals of uncertain sentience such as "fish, invertebrates such as crustaceans, snails and insects", it is a "requirement of a morally decent (or virtuous) person that she at least pays attention to and is cautious regarding the possibly morally relevant aspects of such animals".

Environmental ethics

Kai Chan advocates an environmental ethic, a form of ethical extensionism applied to all living beings. He cites "a non-zero probability of sentience and consciousness" and argues that "we cannot justify excluding beings from consideration on the basis of uncertainty of their sentience".

Ethics of artificial intelligence

Nick Bostrom and Eliezer Yudkowsky argue that if an artificial intelligence is sentient, then it is wrong to inflict unnecessary pain on it, as it is wrong to inflict pain on an animal, unless there are "sufficiently strong morally overriding reasons to do so". They also propose the "Principle of Substrate Non-Discrimination", which states: "If two beings have the same functionality and the same conscious experience, and differ only in the substrate of their implementation, then they have the same moral status."

AI veganism

AI veganism applies the rules and concepts of veganism to artificial intelligence (AI). The term has been used for the view that people should abstain from using AI because of its effects on people, animals or the environment.

Some AI vegans have compared the use of data without consent to train AI systems with harms inflicted on animals through animal husbandry. They have also compared the environmental effects of animal husbandry with those of AI training and use. On an individual level, some AI vegans argue that both consuming animal products and using AI can harm the consumer or user.

Some people avoid using large language models because they believe the training process is harmful to people or the planet.

Neuroethics

Adam J. Shriver argues for "precise, precautionary, and probabilistic approaches to sentience" and states that neuroscience has different relevance to each. He concludes that basic protections for animals should be guided by the precautionary principle. He also argues that, although neuroscientific evidence is not always necessary to indicate that members of some species require protection, the "ongoing search for the neural correlates of sentience must be pursued in order to avoid harms that occur from mistaken accounts".

See also

References

Further reading

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